Project Description
Whisper Notes

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Project Idea

Whisper Notes is a React-based web application that helps students convert recorded lecture audio into text using the open-source Whisper speech recognition model. Users can upload an audio file, send it to a backend service for transcription, and review the generated text in a structured frontend interface.

Target Users and Main Goals

The main target users are exchange and international students who attend lectures in a foreign language and may have difficulties understanding everything during class. The application helps them turn lecture recordings into readable text, so they can review the content afterwards and use external tools for translation if needed. The main goal is to provide a usable frontend with clear navigation, meaningful interaction, and a realistic transcription workflow.

Core Frontend Features

Must Have

- Multiple views with clear navigation, for example Home, Transcribe, Results, and About
- Audio upload form with validation for formats such as `.mp3`, `.wav`, `.m4a`, or `.flac`
- Asynchronous communication with the backend API for transcription
- Loading, success, and error states during the transcription process
- Results view that displays the generated transcription text clearly
- Responsive layout for desktop and smaller screens

Good to Have

- Copy or download the transcription result as a `.txt` file
- Choose the spoken language or use automatic language detection
- Choose a Whisper model size, such as `tiny` or `base`
- Save a simple transcription history during the current session
- Record audio directly in the browser using the microphone

Planned Backend / API

The project will use a small backend service, most likely built with Python and FastAPI. The frontend will send the uploaded audio file to a `/transcribe` endpoint using a multipart form request. The backend will temporarily store the file, process it with Whisper, and return a JSON response containing the transcription text and optional information such as detected language or processing time.

Realistic Project Scope

The project is focused but will cover the main frontend techniques from the course: semantic HTML structure, CSS layout and responsive styling, JavaScript form handling, React components, state management, routing, and asynchronous API communication. Advanced features such as user accounts, cloud storage, real-time transcription, and large-scale deployment are outside the scope of this course project.

AI tools were used to support the editing of this document, grammar improvement, and the preparation of the LaTeX code.